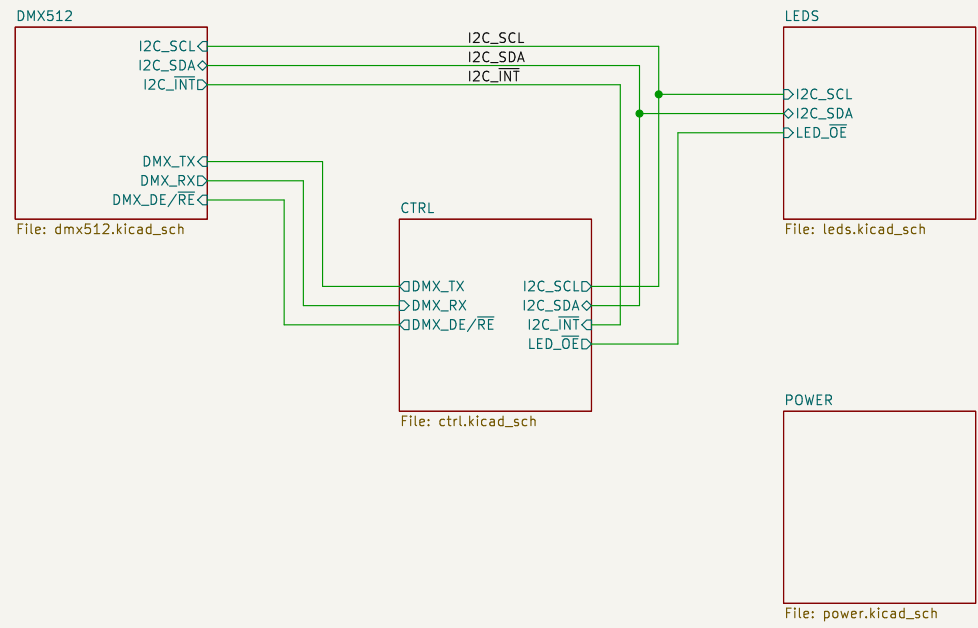


LED Beacon

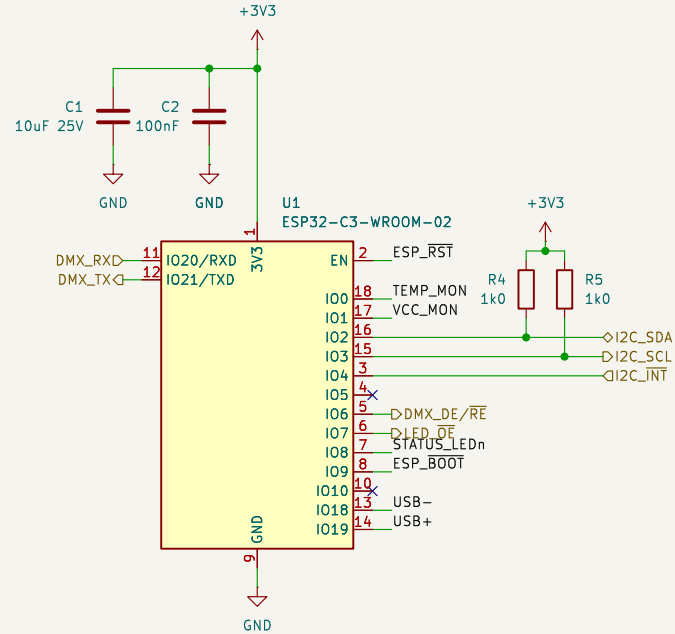
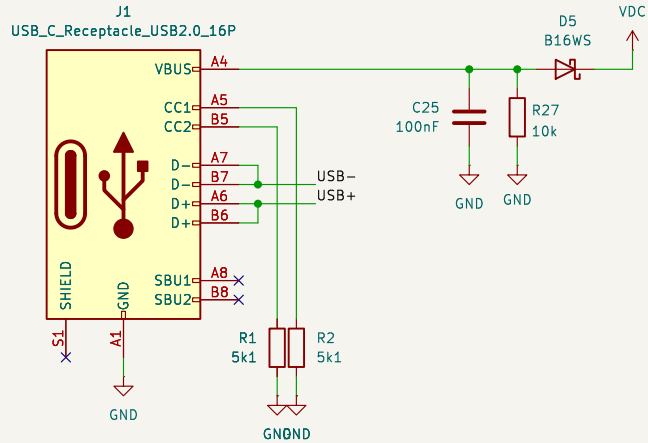


- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

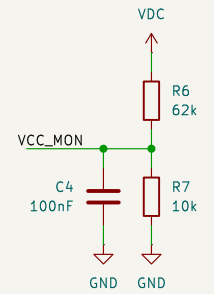
Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 1/5

Controller

USB Programming

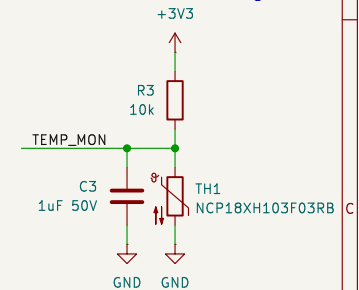


VCC Monitoring

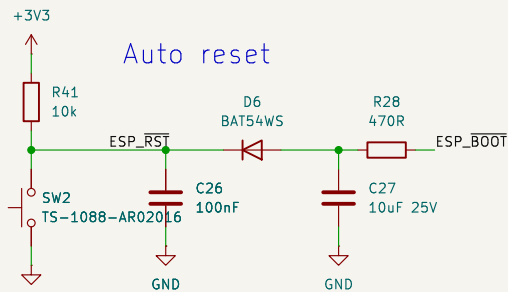


Voltage Monitor
 V_{max} is 17.5V
 $V_{adc} \times ((43k\Omega + 10k\Omega) + 10k\Omega) = VCC$
 $V_{adc} \times 5.3 = VCC$
 $1lsb = 4.271\mu V$

Temperature Monitoring

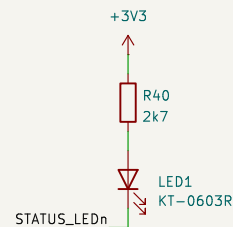


Auto reset



Short press resets the ESP32
 Long press resets the ESP32
 and puts it in bootloader mode

Status LED



IO8 needs to be held high to boot
 so LED & resistor are high side.
 Set STATUS_LEDn to 0V to light LED.

Wires & Bits

Sheet: /CTRL/
 File: ctrl.kicad_sch

Title:

Size: A4
 Date: 2026-04-10
 KiCad E.D.A. 9.0.9

Rev: 0
 Id: 2/5

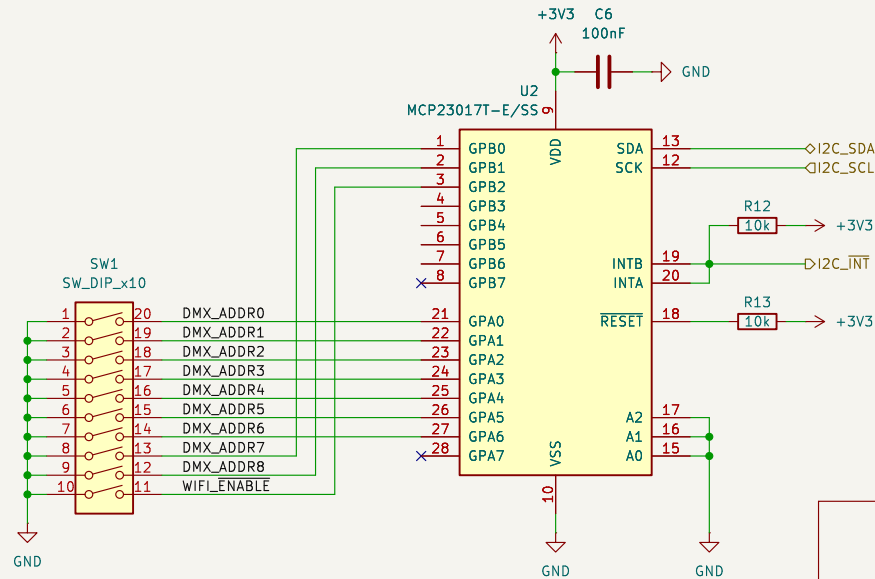
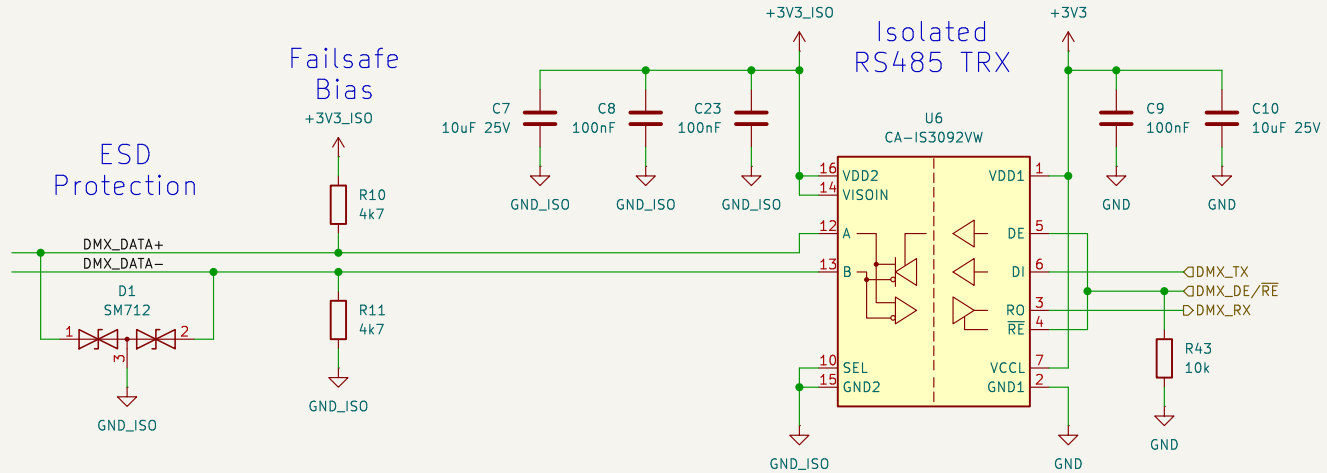
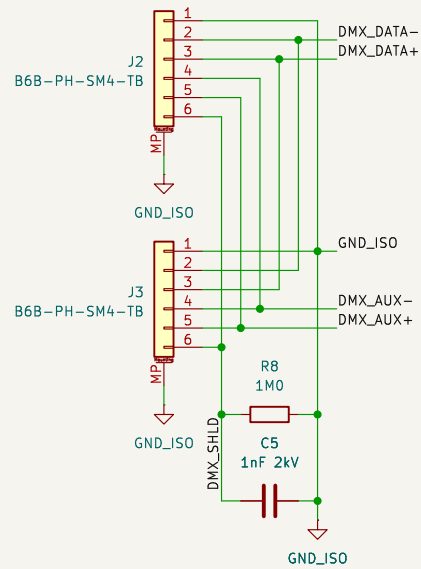
DMX512

Failsafe Bias
+3V3_ISO

ESD Protection

Isolated RS485 TRX

To XLRs



I²C Addr: 0x20
MCP23017 has a fault with GPA7 & GPB7.
The can corrupt an I²C transaction if
they change state during it.

Wires & Bits

Sheet: /DMX512/
File: dmx512.kicad_sch

Title:

Size: A4
Date: 2026-04-10
KiCad E.D.A. 9.0.9

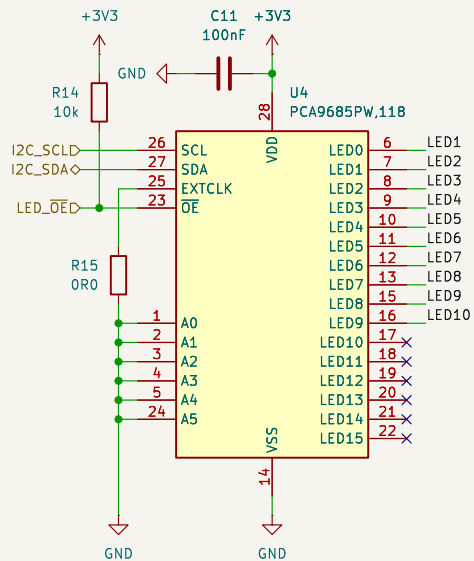
Rev: 0
Id: 3/5

LED Driver

LED Enable

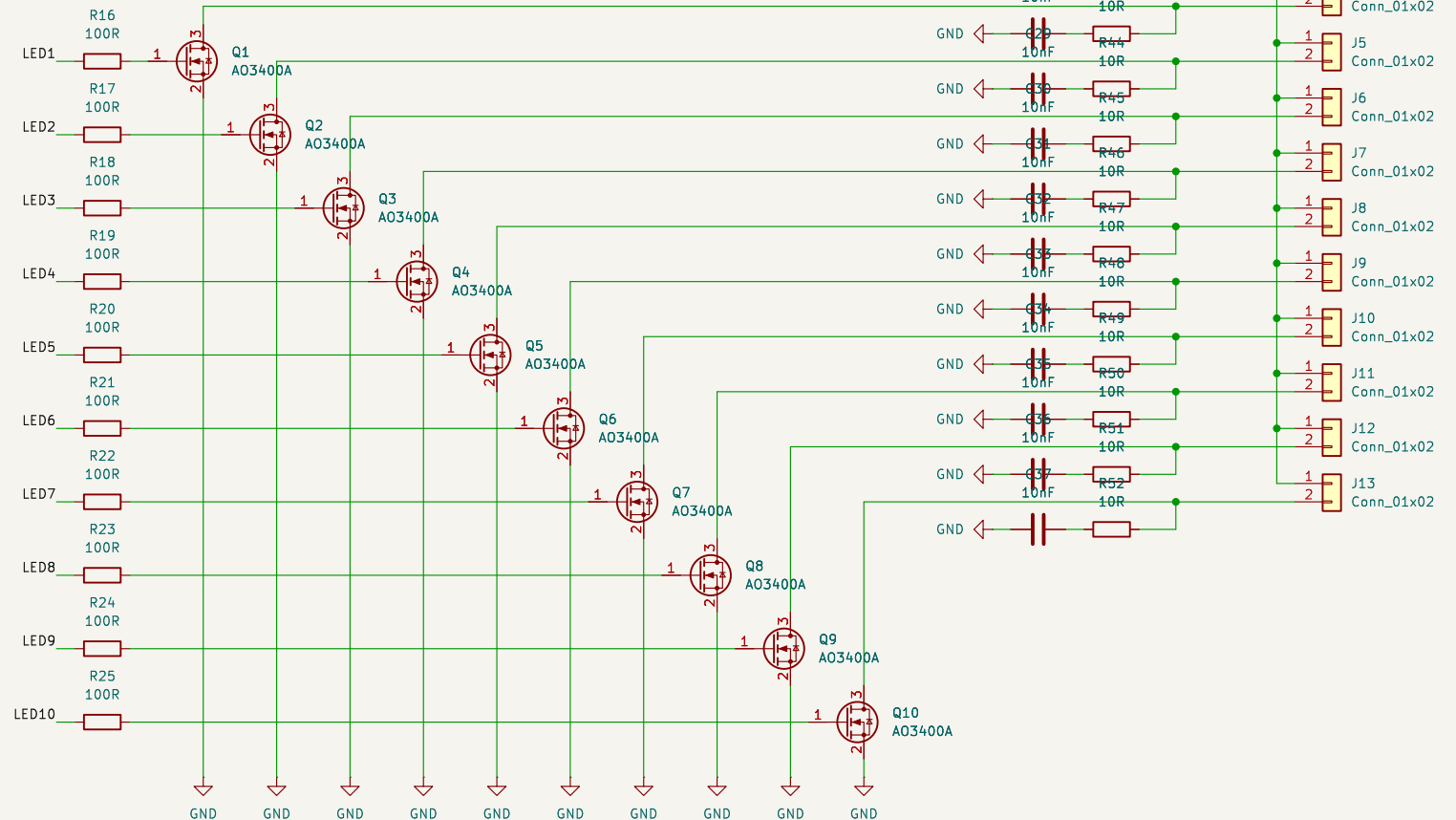
LED Sticks

I2C → PWM



I2C Addr: 0x40

A string of 6x LEDs is 6.4Vf @ 0.3A



Wires & Bits

Sheet: /LEDS/
File: leds.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.9

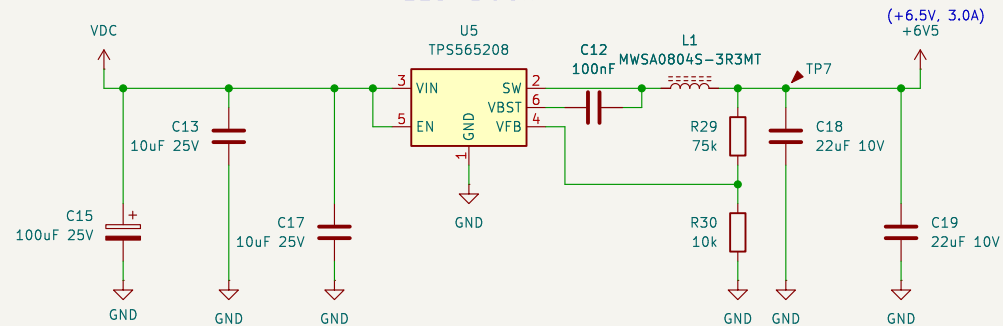
Date: 2026-04-10

Rev: 0

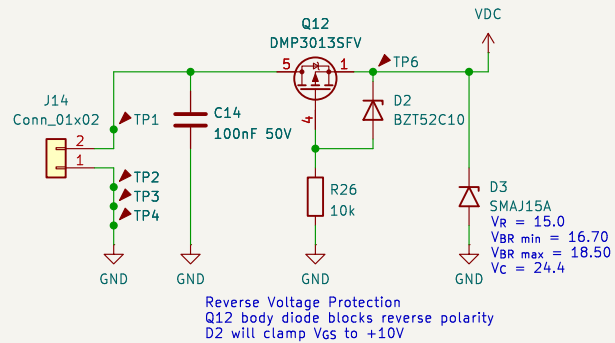
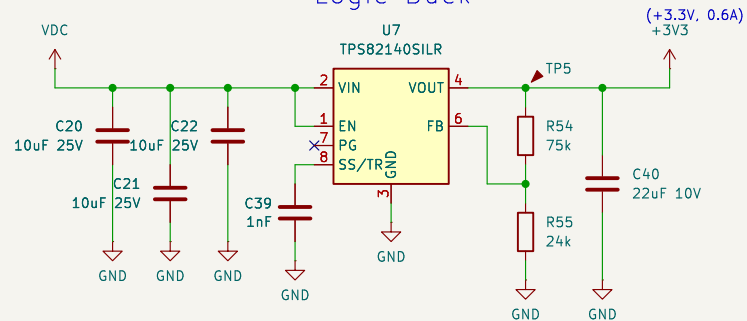
Id: 4/5

Power Supplies

LED Buck



Logic Buck



EMC ferrites /CMC?

Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title:

Size: A4
Date: 2026-04-10
KiCad E.D.A. 9.0.9

Rev: 0
Id: 5/5